HARD · ALGEBRA · ANSWER KEY

Systems of two linear equations in two variables

02

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ANSWER KEY · SAT QUESTION BANK

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Q1

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Answer not provided in source data — see rationale below.

The correct answer is $\frac{1}{2}$. A system of two linear equations has no solution when the graphs of the equations have the same slope and different y-intercepts. Each of the given linear equations is written in the slope-intercept form, $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept of the graph of the equation. For these two linear equations, the y-intercepts are $(0,8)$ and $(0,10)$. Thus, if the system of equations has no solution, the slopes of the graphs of the two linear equations must be the same. The slope of the graph of the first linear equation is $\frac{1}{2}$. Therefore, for the system of equations to have no solution, the value of c must be $\frac{1}{2}$. Note that 1/2 and .5 are examples of ways to enter a correct answer.

Q2

B

B. $\frac{1}{3}x = 4y$

Choice B is correct. A system of two linear equations in two variables, x and y, has no solution when the lines in the xy-plane representing the equations are parallel and distinct. Two lines are parallel and distinct if their slopes are the same and their y-intercepts are different. The slope of the graph of the given equation, $3x = 36y - 45$, in the xy-plane can be found by rewriting the equation in the form $y = mx + b$, where m is the slope of the graph and $0, b$ is the y-intercept. Adding 45 to each side of the given equation yields $3x + 45 = 36y$. Dividing each side of this equation by 36 yields $\frac{1}{12}x + \frac{5}{4} = y$, or $y = \frac{1}{12}x + \frac{5}{4}$. It follows that the slope of the graph of the given equation is $\frac{1}{12}$ and the y-intercept is $0, \frac{5}{4}$. Therefore, the graph of the second equation in the system must also have a slope of $\frac{1}{12}$, but must not have a y-intercept of $0, \frac{5}{4}$. Multiplying each side of the equation given in choice B by $\frac{1}{4}$ yields $\frac{1}{12}x = y$, or $y = \frac{1}{12}x$. It follows that the graph representing the equation in choice B has a slope of $\frac{1}{12}$ and a y-intercept of $0, 0$. Since the slopes of the graphs of the two equations are equal and the y-intercepts of the graphs of the two equations are different, the equation in choice B could be the second equation in the system.

Choice A is incorrect. This equation can be rewritten as $y = \frac{1}{4}x$. It follows that the graph of this equation has a slope of $\frac{1}{4}$, so the system consisting of this equation and the given equation has exactly one solution, rather than no solution.

Choice C is incorrect. This equation can be rewritten as $y = \frac{1}{12}x + \frac{5}{4}$. It follows that the graph of this equation has a slope of $\frac{1}{12}$and a y-intercept of $0, \frac{5}{4}$, so the system consisting of this equation and the given equation has infinitely many solutions, rather than no solution.

Choice D is incorrect. This equation can be rewritten as $y = \frac{1}{36}x + \frac{5}{4}$. It follows that the graph of this equation has a slope of $\frac{1}{36}$, so the system consisting of this equation and the given equation has exactly one solution, rather than no solution.

Q3

87

The correct answer is 87. It's given that in August, the car dealer completed 15 more than 3 times the number of sales the car dealer completed in September. Let x represent the number of sales the car dealer completed in September. It follows that $3x + 15$ represents the number of sales the car dealer completed in August. It's also given that in August and September, the car dealer completed 363 sales. It follows that $x + 3x + 15 = 363$, or $4x + 15 = 363$. Subtracting 15 from each side of this equation yields $4x = 348$. Dividing each side of this equation by 4 yields $x = 87$. Therefore, the car dealer completed 87 sales in September.

Q4

B

B. -5

Choice B is correct. Subtracting the second equation, $4x - 8y = 9$, from the first equation, $7x - 5y = 4$, results in $(7x - 5y) - (4x - 8y) = 4 - 9$, or $7x - 5y - 4x + 8y = 5$. Combining like terms on the left-hand side of this equation yields $3x + 3y = -5$.

Choice A is incorrect and may result from miscalculating $4 - 9$ as -13. Choice C is incorrect and may result from miscalculating $4 - 9$ as 5. Choice D is incorrect and may result from adding 9 to 4 instead of subtracting 9 from 4.

Q5

B

B. $x+3y=-20$

Choice B is correct. A system of two linear equations written in standard form has no solution when the equations are distinct and the ratio of the x-coefficient to the y-coefficient for one equation is equivalent to the ratio of the x-coefficient to the y-coefficient for the other equation. This ratio for the given equation is 2 to 6, or 1 to 3. Only choice B is an equation that isn't equivalent to the given equation and whose ratio of the x-coefficient to the y-coefficient is 1 to 3.

Choice A is incorrect. Multiplying each of the terms in this equation by 2 yields an equation that is equivalent to the given equation. This system would have infinitely many solutions. Choices C and D are incorrect. The ratio of the x-coefficient to the y-coefficient in $6x-2y=0$ (choice C) is -6 to 2, or -3 to 1. This ratio in $6x+2y=10$ (choice D) is 6 to 2, or 3 to 1. Since neither of these ratios is equivalent to that for the given equation, these systems would have exactly one solution.

Q6

D

D. 71

Choice D is correct. Multiplying the second equation in the given system by 2 yields $\frac{10}{2}x + 12y = 46$. Adding this equation to the first equation in the system yields $\frac{7}{2}x + 6y + \frac{10}{2}x + 12y = 25 + 46$, which is equivalent to $\frac{7}{2}x + \frac{10}{2}x + 6y + 12y = 25 + 46$, or $\frac{17}{2}x + 18y = 71$. Therefore, the value of $\frac{17}{2}x + 18y$ is 71.

Choice A is incorrect. This is the value of x, not the value of $\frac{17}{2}x + 18y$.

Choice B is incorrect. This is the value of y, not the value of $\frac{17}{2}x + 18y$.

Choice C is incorrect. This the value of $\frac{7}{2}x + 6y + \frac{5}{2}x + 6y$, or $6x + 12y$, not the value of $\frac{17}{2}x + 18y$
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Q7 .2857

Accept also: 2/7

The correct answer is $\frac{2}{7}$. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions if and only if the two linear equations are equivalent. Multiplying each side of the first equation in the system by $\frac{35}{4}$ yields $\frac{35}{4}\frac{2}{5}x + \frac{7}{5}y = \frac{35}{4}\frac{2}{7}$, or $\frac{7}{2}x + \frac{49}{4}y = \frac{5}{2}$. Since this equation is equivalent to the second equation and has the same right side as the second equation, the coefficients of x and y, respectively, should also be the same. It follows that $g = \frac{7}{2}$ and $k = \frac{49}{4}$. Therefore, the value of $\frac{g}{k}$ is $\frac{\frac{7}{2}}{\frac{49}{4}}$, or $\frac{2}{7}$. Note that 2/7, .2857, 0.285, and 0.286 are examples of ways to enter a correct answer.

Q8

C

C. I and II

Choice C is correct. A system of two linear equations in two variables has at least one solution if the equations represent the same line or lines that intersect at exactly one point. Statement I gives the equation $3x + 13.5y = 10.5$. Multiplying both sides of the given equation, $2x + 9y = 7$, by $\frac{3}{2}$ yields $3x + 13.5y = 10.5$, so the equation in statement I is equivalent to the given equation. Therefore, the system formed by these two equations represents the same line and has infinitely many solutions. Statement II gives the equation $3x - 13.5y = 10.5$. Adding the right- and left-hand sides of $3x + 13.5y = 10.5$ and $3x - 13.5y = 10.5$ yields $3x + 13.5y + 3x - 13.5y = 10.5 + 10.5$, or $6x = 21$. Dividing both sides of this equation by 6 yields $x = \frac{21}{6}$. Therefore, the lines represented by these equations intersect at exactly one point, where $x = \frac{21}{6}$, and the system formed by the given equation and the equation in statement II has one solution. It follows that both the equation in statement I and the equation in statement II could be the other equation in a system of equations with at least one solution.

Choice A is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

Choice B is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

Choice D is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

Q9 3.5

Accept also: 7/2

The correct answer is $\frac{7}{2}$. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the xy-plane are distinct and parallel. Two lines represented by equations in standard form $Ax + By = C$, where A, B, and C are constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation. The first equation in the given system, $\frac{7}{8}y - \frac{5}{8}x = \frac{4}{7} - \frac{7}{8}y$, can be written in standard form by adding $\frac{7}{8}y$ to both sides of the equation, which yields $\frac{14}{8}y - \frac{5}{8}x = \frac{4}{7}$, or $-\frac{5}{8}x + \frac{14}{8}y = \frac{4}{7}$. Multiplying each term in this equation by -8 yields $5x - 14y = -\frac{32}{7}$. The second equation in the given system, $\frac{5}{4}x + \frac{7}{4} = py + \frac{15}{4}$, can be written in standard form by subtracting $\frac{7}{4}$ and py from both sides of the equation, which yields $\frac{5}{4}x - py = \frac{8}{4}$. Multiplying each term in this equation by 4 yields $5x - 4py = 8$. The coefficient of x in the first equation, $5x - 14y = -\frac{32}{7}$, is equal to the coefficient of x in the second equation, $5x - 4py = 8$. For the lines to be parallel, and for the coefficients for x and y in one equation to be proportional to the corresponding coefficients in the other equation, the coefficient of y in the second equation must also be equal to the coefficient of y in the first equation. Therefore, $-14 = -4p$. Dividing both sides of this equation by -4 yields $\frac{-14}{-4} = p$, or $p = \frac{7}{2}$. Therefore, if the given system of equations has no solution, the value of p is $\frac{7}{2}$. Note that 7/2 and 3.5 are examples of ways to enter a correct answer.

Q10 B

B. $-6x + y = 22$

Choice B is correct. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the xy-plane are parallel and distinct. Lines represented by equations in standard form, $Ax + By = C$ and $Dx + Ey = F$, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation, meaning $\frac{D}{A} = \frac{E}{B}$; and the lines are distinct if the constants are not proportional, meaning $\frac{F}{C}$ is not equal to $\frac{D}{A}$ or $\frac{E}{B}$. The given equation, $y = 6x + 18$, can be written in standard form by subtracting $6x$ from both sides of the equation to yield $-6x + y = 18$. Therefore, the given equation can be written in the form $Ax + By = C$, where $A = -6$, $B = 1$, and $C = 18$. The equation in choice B, $-6x + y = 22$, is written in the form $Dx + Ey = F$, where $D = -6$, $E = 1$, and $F = 22$. Therefore, $\frac{D}{A} = \frac{-6}{-6}$, which can be rewritten as $\frac{D}{A} = 1$; $\frac{E}{B} = \frac{1}{1}$, which can be rewritten as $\frac{E}{B} = 1$; and $\frac{F}{C} = \frac{22}{18}$, which can be rewritten as $\frac{F}{C} = \frac{11}{9}$. Since $\frac{D}{A} = 1$, $\frac{E}{B} = 1$, and $\frac{F}{C}$ is not equal to 1, it follows that the given equation and the equation $-6x + y = 22$ are parallel and distinct. Therefore, a system of two linear equations consisting of the given equation and the equation $-6x + y = 22$ has no solution. Thus, the equation in choice B could be the second equation in the system.

Choice A is incorrect. The equation $-6x + y = 18$ and the given equation represent the same line in the xy-plane. Therefore, a system of these linear equations would have infinitely many solutions, rather than no solution.

Choice C is incorrect. The equation $-12x + y = 36$ and the given equation represent lines in the xy-plane that are distinct and not parallel. Therefore, a system of these linear equations would have exactly one solution, rather than no solution.

Choice D is incorrect. The equation $-12x + y = 18$ and the given equation represent lines in the xy-plane that are distinct and not parallel. Therefore, a system of these linear equations would have exactly one solution, rather than no solution.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q1 — the source CB data omitted a structured key for these items, so the answer slot shows –. The worked solution in the rationale below each entry contains the correct value.

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